

Extending the KRC–Davinci Interface

Subsurface solar energy deposition, temperature-dependent emissivity, adaptive meshing, and ice thermal conductivity

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Abstract

This note documents four additions made to the Davinci interface that drives KRC (`krc_davinciinterface.dvrc`, function `krc()`, and its helper `krc_evalN1`, originally written by S. Piqueux): (i) distribution of absorbed solar energy within the subsurface rather than at the surface alone, (ii) a temperature-dependent surface emissivity, (iii) an adaptive first-layer thickness that keeps the near-surface mesh consistent with (i), and (iv) temperature-dependent thermal conductivity and density for water ice. The first two correspond to new KRC input cards (17 and 18); the mesh adaptation is interface-side bookkeeping required by the first; the ice properties feed the existing T -dependent conductivity card (12) through a material-property lookup. Prescribing MCD/PCM atmospheric fluxes in KRC (`UserPrescribed_Fluxes`, card 8/26) is a fifth, already-documented change: it is only summarized here, with pointers to the dedicated design document.

1 Context

`krc_davinciinterface.dvrc` builds a KRC input deck from high-level physical parameters, sizes the default vertical mesh (`krc_evalN1`), and manages external flux and albedo files. It is the layer between physically meaningful arguments (thermal inertia, albedo, orbital elements, ...) and the fixed-format card deck that the KRC Fortran executable reads (see `docs/flux_tables.md` and the two new-feature notes, `docs/solar_penetration_in_ground.md` and `docs/emissivity_temperature_dependant.md`, in the KRC repository). The four modifications below extend both the KRC side (two new card types) and the interface side (argument list of `krc()`, mesh sizing logic, and a material-property helper).

2 Subsurface solar energy deposition (RADGND / EFOLD_RADGND)

By default KRC deposits all absorbed insolation $S_0(1 - A)$ at the surface, which appears as a source term in the surface radiative balance. For optically translucent materials (e.g. water ice), a fraction of that energy is instead absorbed in depth. RADGND switches on a volumetric deposition following a Beer–Lambert-type exponential profile with e-folding length EFOLD_RADGND (in metres):

$$\frac{dQ}{dz} \propto \exp\left(-\frac{z}{\ell}\right), \quad \ell \equiv \text{EFOLD_RADGND}, \quad (1)$$

and removes the corresponding term from the surface boundary condition, which then reduces to $\varepsilon \sigma T_{\text{surf}}^4 = -k \partial T / \partial z$ (no $S_0(1 - A)$ term at $z = 0$). On the interface side this is exposed as two new `krc()` arguments, translated into KRC card 17:

Argument	Meaning	KRC card
RADGND	enable subsurface deposition (boolean)	17 1 <0/1> 'RADGND' /
EFOLD_RADGND	e-folding length ℓ [m]	17 2 <val> 'EFOLD_RADGND' /

! A mesh sized only for surface thermal diffusion can be far coarser than ℓ ; Section 4 describes the mesh-side consequence of enabling RADGND. Validation: `Test_effectTivssolarpene`, `Test_constantabslength_vs_cosilength` (incidence-angle scaling of the absorption length), against the analytical reference in `Validation_solarpenetration.dv`.

3 Temperature-dependent surface emissivity (EmisT)

EmisT replaces the constant surface emissivity by a third-order polynomial in temperature, evaluated at each iteration of the surface energy balance:

$$\varepsilon(T) = \text{Emis0} + \text{Emis1}X + \text{Emis2}X^2 + \text{Emis3}X^3, \quad X = \frac{T - 220}{100}, \quad (2)$$

the same normalized-temperature basis already used by the T -dependent conductivity and specific-heat cards (12/13). The result is clamped to $[0, 1]$; a non-positive value aborts the current case (KRC’s blow-up handler), so coefficient sets should be checked against the expected surface temperature range before production use. Interface side, this is card 18:

Argument	Meaning	KRC card
EmisT	enable T -dependent emissivity (boolean)	18 1 <0/1> 'EmisT' /
Emis0–Emis3	polynomial coefficients	18 2..5 <val> 'Emis0'..'Emis3' /

Coefficients are obtained by fitting $\varepsilon(T)$ derived from a Mie/Hapke emissivity model for amorphous and crystalline water ice (`Compute_Fit_Emissivity_4KRC.dvrc`, inputs `emis_amorphous_ice.txt`, `emis_crystalline_ice.txt`); fit quality is checked in `Figures/Figure_EmissIcvsT.pdf` against the tabulated true emissivity for a range of grain radii and absorption coefficients. The combined effect of EmisT and T -dependent conductivity on T_{surf} is checked separately in `combined_effect_emissT_and_conductivity`.

4 Adaptive first-layer thickness (FLAY)

`krc_evalN1` sizes the default vertical mesh (first-layer thickness FLAY, layer count IC2/N1) from the diurnal and seasonal skin depths implied by the surface thermal inertia, density, and specific heat, for three layering cases (homogeneous half-space, fixed-thickness two-layer medium, or an exponential Hayne-type profile). When RADGND is active, the resulting FLAY is additionally capped so that the first mesh element stays fine compared to the solar e-folding length, guaranteeing that the volumetric heating profile of Section 2 is numerically resolved:

$$\text{if RADGND} = \text{T and FLAY2} > \frac{\ell}{3\text{RLAY} \cdot \text{RLAY} \cdot \text{DSD1_m} \cdot (0.5 + (1 - \text{RLAY}))}, \quad \text{FLAY2} \leftarrow \text{that bound}, \quad (3)$$

where `DSD1_m` is the diurnal skin depth of the top material and `RLAY` the layer-thickness growth ratio. Without this cap, a mesh dimensioned for surface thermal diffusion alone can be coarser than ℓ , biasing the in-depth heating profile. Validated in `Test_adaptativeFLAY` by comparing fixed-FLAY and adaptive-FLAY runs at two thermal inertias (33 and $133 \text{ J m}^{-2} \text{ K}^{-1} \text{ s}^{-1/2}$) and checking convergence of T_{surf} .

5 Temperature-dependent ice thermal properties

The material-property helper `Mat_Prop` returns, for a given material, the polynomial coefficients expected by KRC’s T -dependent conductivity and specific heat cards (12/13), on the same normalized basis $X = (T - 220)/100$ as Section 3. The "H20" entry encodes:

- thermal conductivity $k(T)$ after Slack (1980), valid 60 K to 270 K;

- density $\rho(T)$ after Röttger et al. (2012), valid 10 K to 265 K;
- specific heat $C_p(T)$ after Giauque & Stout (1936), valid 15 K to 273 K.

These defaults are derived from, and cross-checked against, an icy-regolith conductivity model reproducing Ferrari & Lucas (`Scripts_reproduceFerrariandLucas`) and initially refit for standalone injection into KRC (`Script_FitIceproperties4KRC`). End-to-end sensitivity of KRC output to the new coefficients (surface temperature and effective thermal inertia, as a function of albedo and stratigraphy) is checked in `Test_newcondinKRC`.

Joint fit with emissivity. Since the conductivity of icy regolith itself depends on emissivity (radiative conductivity term), grain radius and porosity, fitting conductivity and emissivity independently is only an approximation. `Compute_Fit_PropIcyRegolith_4KRC.dvrc` replaces the two separate fits above by a single, self-consistent one (Davinci function `Compute_Fit_ThermophysicalProperties_IcyRegolith`): given a grain radius, porosity, conductivity mixing model (GJKRBB/GB/WBB/WGB) and ice form (amorphous/crystalline), it returns the `Emis0–Emis3` coefficients of Section 3 together with matching `Con0–Con3` and `Sph0–Sph3` coefficients for card 12/13, recomputing emissivity internally when it is temperature-dependent rather than assuming a fixed default. This is the current, production version of this fit; the effect of enabling each T -dependence (conductivity, emissivity, or both) on T_{surf} is checked in `joint_fit_validation`.

6 Summary of new `krc()` arguments

Argument	Purpose	KRC card
<code>RADGND</code>	enable subsurface solar deposition	17 1 <0/1> 'RADGND' /
<code>EFOLD_RADGND</code>	solar absorption e-folding length [m]	17 2 <val> 'EFOLD_RADGND' /
<code>EmisT</code>	enable T -dependent emissivity	18 1 <0/1> 'EmisT' /
<code>Emis0–Emis3</code>	$\epsilon(T)$ polynomial coefficients	18 2..5 <val> 'Emis0'..'Emis3' /
<code>UserPrescribed_Fluxes</code>	prescribed VIS/IR/ATM flux table (MCD/PCM)	8 26 0 '<file>' /

`UserPrescribed_Fluxes` is unchanged by the present work and is documented in full (format, database construction, coupling tutorial) in the companion design document *Prescribing Mars PCM/MCD Atmospheric Fluxes in KRC*; it is listed here only for completeness, since it shares the interface file and the same argument-to-card mechanism as the four additions above.

7 Validation trace

Complete inputs, scripts and figures are archived alongside this note: `KRC_repo_additions/docs/validation/` for the two KRC-side cards (`RADGND`, `EmisT` — Sections 2–3); adaptive FLAY validation sits with it (`Test_adaptativeFLAY`, Section 4, since it is checked through the `RADGND` card). On the interface side, `Davinci_interface/CouplingwithMCD_PCM/` covers `UserPrescribed_Fluxes`, and `Davinci_interface/Improve_IceProperties/` covers the ice thermal properties of Section 5, including the joint fit. A short README in every folder of that archive states its content and how it maps to the sections above.